Evaluating Open-Source Vision-Language Models for Multimodal Sarcasm Detection

Saroj Basnet
George Mason University
Fairfax, VA, USA

Shafkat Farabi
Virginia Tech
Alexandria, VA, USA

Tharindu Ranasinghe
Lancaster University
Lancaster, UK

Diptesh Kanojia
University of Surrey
Guildford, UK

Marcos Zampieri
George Mason University
Fairfax, VA, USA

***Abstract*—Recent advances in open-source vision-language models (VLMs) offer new opportunities for understanding complex and subjective multimodal phenomena such as sarcasm. In this work, we evaluate seven state-of-the-art VLMs - BLIP2, InstructBLIP, OpenFlamingo, LLaVA, PaliGemma, Gemma3, and Qwen-VL - on their ability to detect multimodal sarcasm using zero-, one-, and few-shot prompting. Furthermore, we evaluate the models' capabilities in generating explanations to sarcastic instances. We evaluate the capabilities of VLMs on three benchmark sarcasm datasets (Muse, MMSD2.0, and SarcNet). Our primary objectives are twofold: (1) to quantify each model's performance in detecting sarcastic image–caption pairs, and (2) to assess their ability to generate human-quality explanations that highlight the visual–textual incongruities driving sarcasm. Our results indicate that, while current models achieve moderate success in binary sarcasm detection, they are still not able to generate high-quality explanations without task-specific fine-tuning.**

***Index Terms*—vision and language, multimodal sarcasm, LLMs**

I. Introduction

Sarcasm is a nuanced form of communication where the intended meaning diverges from the literal expression conveying irony or mockery (see Figure 1). Automatic sarcasm detection has been widely studied in texts [1]–[3] and it is an important part of subjective language understanding [4] along with tasks such as sentiment analysis [5] and offensive language identification [6], [7]. Yet, on social media, sarcasm is often achieved by pairing text with images, and the ironic effect arises from the mismatch between the visual and textual objects. Multimodal sarcasm detection (MSD) has attracted increasing attention due to the proliferation of multimedia content on social platforms. A recent comprehensive survey on the topic [8] highlights the complexity of MSD and the necessity for models capable of interpreting cross-modal incongruities.

Vision–language models (VLMs) pre-trained on large image–text corpora exhibit powerful zero- and few-shot abilities across tasks [9]. The state-of-the-art vision-language models (VLMs) have shown promise in handling multimodal data. Models like Flamingo [10] and VILA [11] have demonstrated capabilities in zero-shot and few-shot learning, adapting to new tasks with minimal examples through in-context learning. These models process interleaved sequences of visual and textual data, making them suitable candidates MSD. However, their performance on this task remains largely unexplored. Notably, studies like Lin *et al.* [12] and Yang *et al.* [13] - include MSD within 14 broader multimodal benchmark datasets used to evaluate VLMs. While these studies benchmark models like GPT-4, LLaMA, OpenFlamingo, BLIP2, and InstructBLIP, only minimal sarcasm data is present.

wow. what a great view!

Fig. 1: Example of a sarcastic instance (image and text pair) from the dataset.

In contrast, [14] have incorporated multimodal large language models (MLLMs) in their pipeline, utilizing CLIP and other visual encoders with training to achieve state-of-the-art performance on MSD. Given the nuanced nature of sarcasm and its reliance on context across modalities, it is imperative to assess whether these models can detect and explain sarcasm without extensive fine-tuning [8], [15]. Recent work on Multimodal Sarcasm Explanation (MuSE) [16] requires models to generate human-style explanations for sarcastic image–caption pairs, but relies on supervised fine-tuning and specialized architectures. We instead ask: *Can off-the-shelf open-source VLMs detect and explain multimodal sarcasm purely through in-context prompts?*

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Building on this foundation, our study focuses on benchmarking widely used open-source VLMs for their ability to identify sarcasm without fine-tuning, utilizing several datasets curated specifically for MSD. Beyond classification, we also evaluate their capability of generating coherent textual explanations reasoning solely through prompting.

The paper addresses the following research questions (RQs):

- **RQ1:** How effectively do state-of-the-art open-source VLMs perform on MSD benchmarks using in-context learning?
- **RQ2:** Can state-of-the-art open-source VLMs generate coherent and task-relevant textual explanations for multimodal sarcasm reasoning solely through prompting?

The main contributions of this paper are as follows:

1) We provide a unified in-context framework with prompt templates that integrate images, few-shot examples, and explanation seeds across seven VLMs.
2) We carry out a classification benchmark with systematic zero-, one-, few-shot evaluation on three MSD benchmark datasets (MMSD, MMSD2.0, SarcNet).
3) We evaluate the explanation generation capabilities each model's free-form irony explanation generation.

II. Related Work

Automatic sarcasm detection has been extensively studied due to its challenging nature [17], [18]. Furthermore, sarcasm has been studied as a linguistic phenomenon part of various related NLP tasks such as sentiment analysis [19]–[21], offensive speech identification [22], and humor identification [23]. Most research on this task addresses the identification of sarcasm in texts [17], [24], [25].

Schifanella *et al.* [26] demonstrate that visual modalities often include cues that help models identifying sarcastic intent and since then multiple studies have addressed MSD. Multiple MSD datasets have been crated from social media data [26]–[28]. Among these, MMSD [28] and its improved version MMSD2.0 [29] have become de-facto benchmarks for MSD. Yue *et al.* [30] recently propose a multilingual multimodal dataset for MSD named SarcNet.

Earlier work on MSD focused on extracting visual and textual features with separate encoders (BERT, ResNet, VIT etc.) and designing techniques to aggregate these features. Most feature aggregation techniques rely on the fact that sarcasm is marked by a characteristic semantic incongruity between the text and image. Various techniques such as inter-modal attention [31], graph neural nets [32], [33], and D&R network [34] have been employed to model this relationship at the feature aggregation stage. More recent works have shifted their focus to model the interaction between the modalities early on in the processing pipeline. For example, Wang *et al.* [35] benchmark multimodal models like VisualBERT, LXMERT, ViLBERT etc. that jointly extract features from images and texts to identify sarcasm. Qin *et al.* [29] also take this route by using OpenAI's CLIP architecture.

In recent years, multimodal LLMs such as GPT-4, LLaVA, and others have demonstrated remarkable capabilities in extracting information and analyzing sentiment from multimodal data. These models have achieved state-of-the-art results across a variety of multimodal tasks. However, as noted by Farabi *et al.* [8] in their survey of the field, there remains a significant gap in the literature: LLMs are still underexplored for MSD. This work helps filling this gap presenting a systematic evaluation of open-source LLMs on MSD using in-context learning.

III. Methods

a) Datasets: We utilize three publicly available datasets presented in Table I for multimodal sarcasm detection. For evaluation, we randomly sample 3,000 image-caption pairs from both MMSD2.0 and SarcNet, multilingual - English and Chinese; datasets for binary classification. In parallel, we draw explanations from the Muse dataset to assess the models' ability not only to label but to qualitatively justify their predictions.

- **MuSE** [16]: Contains image-text pairs labeled for sarcasm, notably including explanations for sarcastic instances. Primarily used for positive (sarcastic) examples in our few-shot prompts.
- **MMSD2.0** [29]: A comparatively larger dataset for multimodal sarcasm detection featuring both sarcastic and non-sarcastic examples. Used for sourcing negative (non-sarcastic) examples for few-shot prompts.
- **SarcNet** [30]: SarcNet is a multilingual (English and Chinese) dataset for multimodal sarcasm detection, comprising 3,335 image-text pair samples.

TABLE I: Overview of key characteristics of datasets used in multimodal sarcasm detection, highlighting the distribution of positive (+) and negative (-) examples, availability of explanations, and support for multilingual data.

Dataset	+ examples	- examples	Explanation	Multilingual
Muse	3,510	0	✓	×
MMSD2.0	11,651	12,980	×	×
SarcNet	1,875	1,460	×	✓

b) Models: We benchmark seven open-source VLMs spanning diverse architectures and pretraining regimes:

- **InstructBlip** [36] - FlanT5 large model further instruction-tuned on multi-task vision-language data.
- **Blip2** `2.7B` [37] - frozen image encoder with Q-former and a large language model backbone.
- **OpenFlamingo** `3B` [38] - a lightweight adaptation of Flamingo for open-source environments.
- **LLaVA** `7B` [39] - vision-language alignment via adversarial fine-tuning of LLaMA.
- **PaliGemma** `3B` [40] - multimodal mixture-of-experts model combining visual and textual experts.
- **Qwen-VL** `7B` [41] – Q-aware encoder-decoder with unified multimodal attention.
- **Gemma3** `27B` [42]– instruction-tuned multimodal model that processes both images and text, and broad multilingual support.

c) In-Context Prompting: We apply a unified `global` prompt instruction template across all-shots settings to ensure the consistency in how each model is queried:

Listing 1: Sarcasm Detection Prompt Structure

```
*<global_instruction>*
Example: (zero-, one-, few-shots)
*<image>*
*Caption:<caption> Answer: Yes/No*
*<image>*
**Context:**: {caption}"
Is this sarcastic?
```

- `Zero-shot`: no examples or reference explanations included.
- `One-shot / Few-shot`: we prepend 1 – 3 exemplar triplets (caption, reference explanation, `\Yes."` or `\No."`) drawn from Muse (positive) and MMSD2.0 (negative).

Prompts are truncated/padded to each model's maximum token length; image inputs use each model's native preprocessing pipeline (PIL Image pixel-values or embedded file-paths). However, due to Gemma3's maximum context window, including more than one example in a few-shot prompt would exceed its capacity, so we omit few-shot evaluations for this model.

Listing 2: Explanation Generation Prompt Structure

```
*<Context>:*
*<image>*
**Original Caption**: {caption}"
**Provided Explanation**: {explnation}
**Task Instruction**
```

This study utilized zero-, one-, and few-shot prompting strategies to evaluate Multimodal sarcasm detection. Zero-shot prompting provided only task instructions and the target query, relying on the model's pre-trained or tuned knowledge. One-shot and few-shot approaches provided one or multiple (N= 1, 3 pairs used for asset building) complete examples, respectively, within the prompt to enable in-context learning. All strategies aimed for a unified output (`Yes/No` classification). Furthermore, we prompted each model to generate a sarcasm explanation using the template in Listing 3. All explanation generation was performed in a strictly zero-shot setting.

d) Reproducibility: All prompt-templates, random-seeds, and data splits are released publicly on Github.[1] Model checkpoints and hyper-parameters are also logged for ease of reproducibility. For further details, see Appendix A

IV. Results

All models were evaluated `as is` with no finetuning or gradient updates. This ensures a fair, out-of-box comparison under strict inference settings. We employed the prompt strategy presented in the previous section. Any generation that failed to adhere precisely to the prescribed `Yes` / `No` format.

[1]https://github.com/pvsnp9/sarcasm_bench

TABLE II: Classification accuracy results of models on the two datasets in one-shot, zero-shot, and few-shot settings. Model and prompting combination sorted by performance.

Dataset	Model	Inference Type	Accuracy
SarcNet	Gemma3	One-shot	0.67
	InstructBlip	Zero-shot	0.67
	InstructBlip	Few-shot	0.66
	Gemma3	Zero-shot	0.66
	Qwen	Few-shot	0.59
	Qwen	One-shot	0.58
	Paligemma	Zero-shot	0.57
	Paligemma	One-shot	0.57
	LlaVA	Zero-shot	0.57
	LlaVA	One-shot	0.56
	Paligemma	Few-shot	0.55
	Openflamingo	One-shot	0.64
	Openflamingo	Few-shot	0.56
	Openflamingo	Zero-shot	0.43
	Blip2	Few-shot	0.44
	Blip2	Zero-shot	0.43
	Blip2	One-shot	0.43
	Qwen	Zero-shot	0.43
	LlaVA	Few-shot	0.43
MMSD2	Gemma3	One-shot	0.73
	InstructBlip	Zero-shot	0.64
	InstructBlip	Few-shot	0.63
	InstructBlip	One-shot	0.63
	Paligemma	One-shot	0.57
	Paligemma	Zero-shot	0.56
	LlaVA	Few-shot	0.51
	Qwen	One-shot	0.51
	Qwen	Zero-shot	0.51
	Qwen	Few-shot	0.50
	Openflamingo	Few-shot	0.49
	LlaVA	One-shot	0.48
	Paligemma	Few-shot	0.48
	Openflamingo	One-shot	0.47
	LlaVA	Zero-shot	0.46
	Blip2	Few-shot	0.53
	Blip2	Zero-shot	0.53
	Blip2	One-shot	0.52
	Openflamingo	Zero-shot	0.52
	Gemma3	Zero-shot	0.60

A. Classification Results

Table II presents the results of all models and prompting strategies on the two datasets (SarcNet and MMSD2.0) in terms of accuracy. The table reveals a clear performance spectrum across the seven multimodal models we evaluated on the two datasets under zero, one, and few-shot prompting. Overall, Gemma3 consistently achieved the strongest result with 0.67 accuracy with one-shot prompting on SarcNet and maintained robust performance across MMSD2.0 with 0.73. Instruction tuned models such as InstuctionBLIP also performed competitively, attaining 0.67, on SarcNet dataset and 0.64 on MMSD2.0 dataset.

We observe that few-shot prompting does not bring performance improvements over one-shot and few-shot prompting strategies. As discussed a recent survey [8], sarcasm detection requires more than pattern matching from a small number of demonstrations. Even well-chosen, carefully-curated examples cannot fully convey the deep world knowledge and fine-grained image–text relationships that underlie ironic intent. Sometimes, adding a single or few demonstrations often introduces noise rather than informative context in the prompts. The models struggle to generalize the nuanced contradiction from just one or

a few examples example, and strict context-window limits force truncation of prompt content. Consequently, simply appending additional shots fails to provide sufficient signal to improve performance on this inherently challenging task.

Overall, while generic vision-language models struggle with the subtlety of sarcasm, instruction tuned architectures like Gemma3 and InstructionBLIP stand out for their robust zero- and one-shot performance. These results highlight the importance of model pretraining, instruction-tuning, and prompt design when tackling high-level reasoning tasks such as multimodal sarcasm detection.

B. Explainability: Generating Explanations

We compute CLIPScore [43], [44] to assess image-text alignment. We evaluate Δ-CLIPScore to quantify the improvement in CLIP alignment from caption to explanation. We introduce the delta CLIP score (ΔCLIP) defined as:

$$\Delta\text{CLIP} = \text{CLIP}(IMG, G_{\text{exp}}) - \text{CLIP}(IMG, B_{\text{exp}})$$

where G_{exp} models explanation, and B_{exp} dataset explanation. Higher Δ CLIP score indicates stronger improvements in grounding. This follows the general idea of "delta" comparative metrics used in other multimodal contexts [45] .

We computed the score over 3,000 samples and present the results in Table III. Both mean and variance are reported to reflect central tendency and stability across the evaluation set.

When prompted to articulate the visual–textual incongruity on the Muse dataset, the seven VLMs exhibit widely varying explanation quality. Even on a dataset of exclusively sarcastic examples with explanations, most vision–language models struggle to reproduce those rationales. Among the seven models we evaluated, Salesforce's BLIP2 achieves the second best overlap with the reference text and a positive embedding-level similarity to the image, making it the clear leader. Instruction-tuned variants like InstructBLIP and OpenFlamingo occupy a middle tier: they generate plausible, partially overlapping explanations but display mixed performance in grounding those explanations back to the image.

TABLE III: Mean and Variance of Δ-CLIPScore Across Models for Explanation Generation

Model	Mean	Variance
Llava	1.966	27.315
Blip2	0.831	25.532
Paligemma	0.757	16.234
Instructblip	0.583	27.749
Gemma3	-2.063	46.481
OpenFlamingo	-1.750	11.526
Qwen	-7.143	25.515

LLaVA stands out for producing the most visually grounded explanations, as measured by CLIP alignment, even though its raw text-overlap scores remain modest. At the other extreme, both Qwen-VL and Paligemma barely exceed a random baseline in any metric, and Google's Gemma3 fails to generate any measurable overlap or alignment with the reference explanations. Finally, every model scores zero on our VQA-style metric, underscoring that none can consistently answer a direct "why" question about the image–caption contrast. These results highlight that only specialized architectures like BLIP2 and LLaVA can approximate human-style sarcasm explanations without further fine-tuning.

While Gemma3 achieves the strongest zero- and one-shot classification accuracy, yet it fails entirely on our text-overlap and embedding-alignment metrics for explanations. Conversely, Llava, one of the weakest classifiers, failing to detect sarcasm, nevertheless generates the most faithful, human-like explanations. Qwen-VL, despite its strong in-context classification performance, produces essentially no overlap with reference explanations.

Finally, we note that the models that excel at binary sarcasm classification are not the same ones that produce high-quality explanations. Models like Gemma3 and Qwen-VL are instructed and tuned primarily for discriminative "Yes/No" decisions, and chat-style generation, so they excel at binary classification. In contrast, high-quality explanations require a generative, VQA-style architecture with deep image–text grounding. Thus, it is the training objective and model design, not only size, that drive this divergence: classification-focused models label effectively, whereas VQA-tuned models produce more faithful, image-grounded rationales. This divergence underscores that task-specific fine-tuning or architectural adaptations may be needed to bridge classification accuracy and explanatory quality in multimodal sarcasm.

V. Conclusion and Future Work

This study provides an evaluation of seven open-source VLMs for MSD and multimodal sarcasm explanation generation using in-context learning. Our experiments addressed two core research questions: **RQ1:** How effectively do state-of-the-art open-source VLMs perform on MSD benchmarks using in-context learning? and **RQ2:** Can state-of-the-art open-source VLMs generate coherent and task-relevant textual explanations for multimodal sarcasm reasoning solely through prompting?

Our findings reveal that, instruction-tuned models such as Gemma3 demonstrate strong performance in binary sarcasm classification; particularly in zero- and one-shot scenarios - answering **(RQ1)**. Our findings also indicate that none of the evaluated models achieves both high classification accuracy and high-quality explanation generation within the same architecture. Models like BLIP2 and LLaVA, despite their lower classification scores, consistently generate explanations that more closely align with human-written rationales. This divergence underscores a critical challenge in multimodal sarcasm: the ability to discriminate sarcasm does not necessarily translate to the capacity for nuanced, interpretable explanation - answering **(RQ2)**.

We observe that increasing the number of in-context examples (few-shot) does not yield gains in sarcasm detection performance. This suggests that current VLMs may require more than simple demonstration-based prompting to generalize the intricate patterns of multimodal irony. In explanation generation, models frequently struggle with both the surface form and

semantic adequacy, highlighting the gap between discriminative and generative capabilities in existing architectures.

In future work, we will focus on bridging the gap between classification and explanation. This includes developing hybrid training objectives that jointly optimize for both tasks (e.g., multi-task learning [46]). We are experimenting with chain-of-thought (CoT) and multi-stage prompting to elicit richer model reasoning, and fine-tuning on datasets that combine sarcasm detection with human-written explanations. Extending model context windows and integrating RAG techniques or external knowledge could further enhance the models' ability to capture the subtle contextual cues inherent in sarcasm. Finally, we would also like to expand benchmarks to include adversarial and multilingual examples, as well as incorporating human-in-the-loop evaluations.

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Appendix

A. Experimental Details

The models used in our experiments are open-source and sourced from the Hugging Face model repository. All models are used "out-of-the-box" (no gradient updates), loaded at their public checkpoints, and evaluated under identical hardware and preprocessing conditions. We utilized the transformers library to load both the models and their corresponding processors. Most models were loaded in 8-bit precision with FlashAttention enabled to optimize memory and computation efficiency. During inference, we used each model's default tokenizer and processing pipeline for input encoding, text generation, and output parsing.

For the few-shot learning setting (1–3 examples), we randomly sampled positive examples from the MUSE dataset—since it contains explanations—and negative examples from the MMSD2.0 dataset. We then constructed a unified prompt for few-shot learning, as illustrated in Listing 1. The generated outputs were parsed using regular expressions, typically matching responses that begin with "Yes" or "No". Outputs that did not match the parsing pattern were logged but excluded from the reported scores in Table II.

Compute Resources: All experiments were conducted on the high-performance computing cluster. We utilized a single compute node equipped with two NVIDIA A100 GPUs (40GB/80GB each), supported by 20GB and 40GB of system RAM. The approximate processing time for each model to handle a batch of 3000 examples (with a batch size of 1) ranged from 2 hours to 2 days, depending on the model and the task.

Global Prompt: This can be found in in `params` file on the GitHub repository made available after acceptance of this paper.

Listing 3: Unified Instruction for MSD

```
"You are an image-caption sarcasm
    detector.\n"
"Sarcasm can arise when the text means
    something different than the image.\n
    "
"For each input, decide whether the
    caption is used sarcastically given
    the image.\n"
"- If sarcastic, output exactly:\n"
"1. Yes"
"\n2. On the next line, A single non-
    sarcastic rephrase of the caption (no
     extra text)\n"
"- If not sarcastic, output exactly:\n"
"1. No"
"\n\nIMPORTANT:\n"
"- Use capital "Y" in Yes and capital "N"
     in  N o nothing  else (no punctuation
    , no bullets).\n"
"DO NOT repeat or echo any part of this
    instruction or the caption.\n"
"Your entire response must be either:'No'
     or 'Yes' \n <rephrase>'\n"
```

These paradigms were adapted for each model's specific input requirements:

- **LLaVA**: Employed a chat format `[INST] <..> [/INST]` (`USER`/`ASSISTANT`) with `<image>` placeholders; the processor handled multiple PIL images (examples + target) corresponding to tokens in a single prompt.
- **OpenFlamingo**: Required pre-processing images into tensors. A single text prompt interleaved `<image>` and text segments (separated by `<|endofchunk|>`), while the corresponding image tensors were passed separately as the `vision_x` input.
- **InstructBlip & Blip2**: Processed a single target image per inference. Few-shot examples were provided textually within the prompt string as Q&A pairs preceding the final query.
- **PaliGemma**: `<image>` placeholders within a single, interleaved prompt string; the processor handled multiple PIL images corresponding to tokens.
- **Qwen-VL** Embedded image file paths directly within the text prompt using `<img>path/to/image.jpg</img>` tags; the tokenizer handled image reference and loading internally. Few-shot examples also used this path embedding format within the prompt prefix.
- **Gemma3**: The inputs were structured in a chat-style format comprising an instruction, a user query, and an image represented as a PIL object. In the few-shot setting, demonstration examples were appended directly before the user's query to provide context for the model.

Inference and Parsing For each prompt, we generate a single output sequence with beam search (beam size=3) and a maximum of 100-256 generated tokens; for most models. We then parse the model's reply via a lightweight regex to extract.

`Classification`: `Yes` → sarcastic `No` → non-sarcastic
`Explanation`: A VLM generated explanation for sarcastic image-caption pairs.

Any unparsable or generic fallback (*"I'm not trained..."*) is logged as an abstention. The unexpected results do not contribute to the final result.

B. Hyperparameters

TABLE IV: Inference-time hyperparameters

Hyperparameter	Value
Batch size	1
Example size	0–3
Max tokens	100–256
Num beams	3
Quantization	FP8, FP16
Flash attention	True
Input padding	True
Input truncation	True
Temperature	0.7

The hyperparameters were selected to balance model compatibility and inference efficiency; for instance, Gemini, we limited our experiments to zero- and one-shot settings to avoid truncation of longer input sequences, thereby preserving the full contextual information necessary for reliable evaluation.